Live Game Operations Rooms

Approach

3 hour goal

- 1h Thinking
- 1h Coding
- 1h Testing/Deliverables

3 hour goal

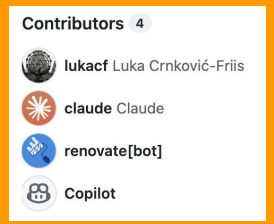
1h Coding hinges on plug n play tools (no bugs, well documented)

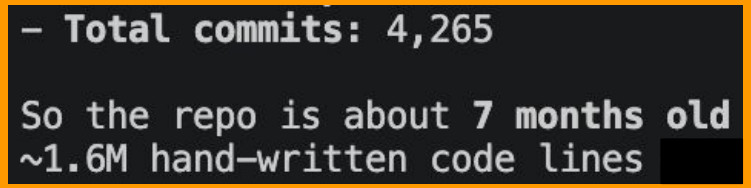
- Assignment requires https://github.com/lukacf/meerkat

CAUTION

Flagged in first 10 minutes

meerkat has a **single** (human) **contributor** and is **largely vibe** coded





➔ In prod, I would recommend frameworks with a community and/or long term guarantees

> what are open source and long term alternatives to using meerkat + meerkat-mob?

If the identity-first multi-agent team concept (Meerkat's core differentiator) is what you want to preserve, these are the closest matches:

- **CrewAI** — the most direct conceptual analog to Meerkat's "mobs": role-based, reusable agent teams with defined goals; the most approachable option for building agent teams, open source with an enterprise tier.langfuse+1
- **AWS Multi-Agent Orchestrator** — Bedrock-native routing between multiple agents, free SDK, good fit if standardizing on AWS.ayautomate
- **AgentScope** — distributed, observable multi-agent framework from Alibaba Cloud, built for high-throughput scenarios.ayautomate
- **Microsoft Conductor** — described as "the most credible long-term bet" among newer multi-agent CLI orchestrators, MIT-licensed, using YAML-defined workflows with a web dashboard, built on GitHub Copilot SDK and Claude.

3 hour goal

1h Coding hinges on plug n play tools (no bugs, well documented)

1. Fallback to Figma/Excalidraw - not acceptable vs AI-skeptic stakeholders
2. Replace meerkat - break meerkat requirement
3. Fill the gaps in meerkat - break time requirement
   - Small extra time > completely different library
   - Stick with existing ecosystem
   - Curiosity
   - One extra to discuss later

Thinking / Coding split

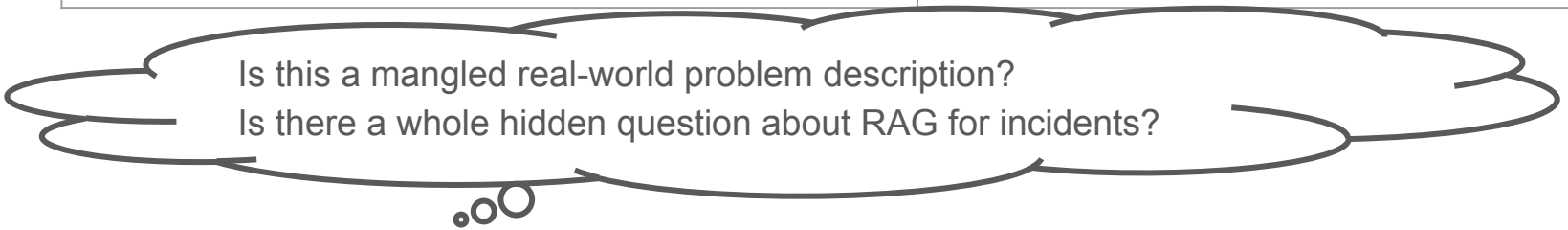
- Demo first
  - ~ 1 full page of requirements
  - Emphasis (“Core requirement”, “Runnable implementation”)
  - Live usage informs UI and architecture decisions

- Architecture second
  - Requirement have 2 lines at the bottom
  - Interview offers opportunity to discuss more
  - Findings from research show simple architectures
    - HolmesGPT uses controler/worker setup
    - Domain specialization (+memory, source-ids, post-mortem, handoff, etc.) hard to devise from project description

Assignment framing

Assignment is specifics-first

Open Context	Specific Implementation
• "Support an SRE team" • "No correct answer" • "Purposeful design" • "Minimal UI"	• 2+ agents minimum • Different responsibilities/context • Real mob collaboration • Stable source ids • Handle late evidence • Preserve earlier evidence • **Account for cost** • Show hypothesis confidence • **Show impact** • Show next action • Show Agent status • **Produce handoff** • No chain-of-thought • Show supporting/conflicting evidence



Candidate enjoyability

Open Implementation	Specific Context
• “Show that team some sort of demo”	• “The incident team responds to incidents with Architecture X and find Y and Z difficult and boring. They have ecosystem X in place, they are accountable for Y, and don’t trust AI.” • Ecosystem uses GTP-5.6 Sol for all agents, here is a key valid for 14 days: `OPENAI_API_KEY=sk-...`

Time breakdown

Honest breakdown - User time 2.4h

Web UI ——— user 53.8m (0.9h) · wall 109.7m (1.8h)
- Scaffold from mockup (UI_v1.png) ——— user 16.3m · wall 43.0m
  - Build approach + first cut ——— user 11.6m · wall 28.6m
  - "proceed" → first render ——— user 4.7m · wall 14.4m
- Panels & correctness ——— user 20.9m · wall 32.5m
  - Nail behavior; show fact-answered tests ——— user 14.9m · wall 21.2m
  - Keep working on UI behavior ——— user 4.2m · wall 4.6m
  - Test status chips (YES/NO/Pending) + falsifier links user 1.8m · wall 6.7m
- Operator-chat UX ——— user 14.1m · wall 29.7m
  - Live status msgs ("creating hypotheses… done") user 12.3m · wall 21.7m
  - Lock chat while a run is in progress ——— user 1.8m · wall 8.0m
- SUB AGENT gag button (rickroll) ——— user 1.6m · wall 3.6m
  - Add the blue button ——— user 1.4m · wall 3.0m
  - Gate it on a pending test being selected ——— user 0.2m · wall 0.6m
- Redeploy & verify ——— user ~0.9m · wall ~0.9m

Prompts ——— user 10.9m (0.18h) · wall 15.6m
- Tighten a gen prompt + bump model to gpt-4.1 ——— user 2.3m · wall 4.2m [session 5]
  "switch to 4.1 and tighten the prompt as well"
- Regenerate / review both generation prompts ——— user 5.6m · wall 5.7m [session 6]
  "give me both prompts again" (hypothesis-gen + test-gen)
- Operator voice — informal, French-interjecting dev — user 3.0m · wall 5.7m [session 6]
  "…the style should be informal, like an old school bearded developer, who often interject in French"

Seed ——— user 5.6m (0.09h) · wall 10.9m
- Wire the incident.json / late-arrival.json schema —— user 5.6m · wall 10.9m [session 6]
  "I added incident.json and late-arrival.json to ace/ that show the correct schema… make sure our system can handle that input schema"

RCA protocol / controller ——— user 71m (1.18h) · wall ~136m
- A. Protocol design (up-front spec) ——— user 14.6m · wall 32.4m
  - "build an RCA system…" initial spec ——— user 1.1m · wall 4.3m
  - blocks→parent edges + falsification rule ——— user 9.8m · wall 11.3m
  - "implement this plan; roster filled on compose up" —— user 3.7m · wall 16.8m (Claude then built it)
- B. Graph-correctness enforcement ——— user ~18m · wall ~34m
  - "single hypothesis has 2 tests" ——— user 7.0m · wall 13.1m
  - "test_runner validating a hyp is unacceptable" ——— user 3.4m · wall 10.0m
  - "diagnosis tests never written if answer in facts?" —— user 2.7m · wall 3.2m
  - "generate_tests gated / 1 test at a time" ——— user 2.8m · wall ~5m
  - "when all hyps fail, regenerate new ones" ——— user 2.1m · wall 4.6m
- C. Drive-model pivot (turn-based → controller-driven) —— user 16.3m · wall 37.4m
  - "Go for Python + switch back to gpt-4o-mini" ——— user 2.6m · wall 9.0m
  - "querying each agent once is not acceptable" ——— user 4.0m · wall 10.8m
  - controller log paste (RPC ready, watching coord.) ——— user 5.4m · wall 9.1m
  - "do both" ——— user 2.2m · wall 3.2m
  - (s6) "redeploy, try controller precise link" ——— user 2.0m · wall 5.3m
- D. Graph-state VERIFICATION (deploy → inspect → confirm) user 22.1m · wall 31.0m

  short messages; the minutes are spent DRIVING the live system, not typing:
  - "process finished, does the graph have expected state?" user 4.9m · wall 7.3m
  - "it finished, does the graph state look correct?" ——— user 5.0m · wall 6.1m
  - "I added this fact, is the graph correct?" ——— user 5.3m · wall 6.1m ← 8-word msg
  - "do it + default facts to incident + cosmetic nodes" —— user 3.1m · wall 6.3m
  - "check it" ——— user 3.8m · wall 4.4m
  - "is the graph state sensible now?" ——— user 0.0m · wall 0.9m

Complete Honest breakdown - Wall clock time 7.1h

Multitask time!

Completely Honest breakdown - meerkat gaps 13h

	FAILURE MODE	DOC VERDICT
A `turn_driven` member answers only its FIRST `console/send` per session; 2nd+ return an empty frame in ~1ms — fix is `respawn_member` before every ask	SILENT	Undoc
The WorkGraph grant is auto-issued by `persistent_state(dir)` — docs imply *you* must issue it	MISLEADING	Misleading
OpenAI 400s the top-level `not` in `workgraph_claim`; the `openai_compatible` adapter won't lower it (also hit as a llama.cpp boolean-schema shim)	SILENT	Undoc · defect
No `tracing` subscriber ⇒ **every meerkat warn/error silently dropped** — see the box below	SILENT	Undoc
Non-orchestrator profiles never auto-spawn — only `coordinator` reconciles on boot; need `ensure_member`	SILENT	Vague
Console binaries bind `127.0.0.1:0` only — no flag/env/config re-binds it; **forced compiling a library host** (whole Rust build saga downstream)	HARD-FAIL	Partial
`build_runtime_decision_state` validators (JWKS keys, audience) crash startup	HARD-FAIL	Partial
A member with `tools.comms=false` is hard-rejected — there is no tool-free member	HARD-FAIL	Undoc
`mobkit/reset_all` takes the whole stack unreachable (HTTP 000) — soft-reset epoch instead	TRAP	Undoc
The `labels` filter is match-ALL (AND) — a multi-label query returns `[]`	SILENT	Undoc
Terminal items are hidden unless you pass `include_terminal:true`	SILENT	Vague
`workgraph/evidence/add` needs its own inner `id` on the evidence object or it fails	TRAP	Undoc

Retrospective

If assignment drops at lunch, presenting the next morning?

- Back to slide 6
  - Meerkat risk was flagged very early
    - Manage expectations
    - Involve stakeholders in path forward
  - Reach out to repo creator to flag landmines early

Image artefacts

Similar incidents

Xxx
yyy

Hypotheses

H1 New deployment
~~H2 DoS~~

Tests

T0 CPU above 80%?
T1 RAM above 99%?

Operator chat

SEND

Hypotheses (supporting | conflicting facts)

(open) H001 (3 | 0) New deployment
(failed) H002 (1 | 2) DoS attack

Tests (related hypo) (outcome)

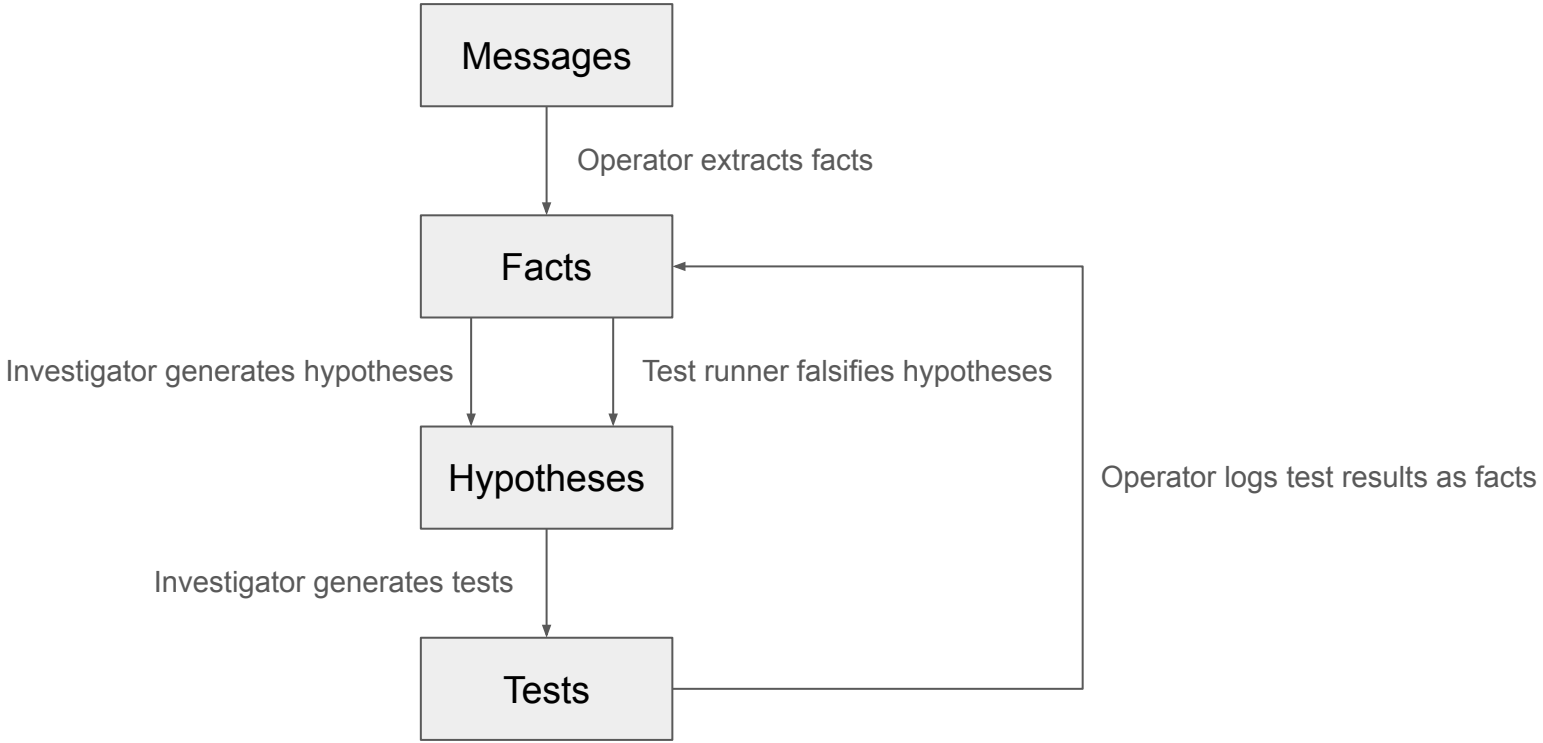
T002 (H002) Is CPU above 80%?
T001 (H001) Is RAM above 99%? (No)

YES NO

Operator chat

User: New Incident xxx, facts xxx, yyy
Operator: Updated workgraph

SEND


Messages
Operator extracts facts
Facts
Investigator generates hypotheses
Test runner falsifies hypotheses
Hypotheses
Operator logs test results as facts
Investigator generates tests
Tests

Notes on deliverables

- “runnable implementation” smuggles overhead and decisions
  - What machine I/you have -> Design guessing
  - Repo is also required, which presumably must produce identical runtime -> CI/CD
  -